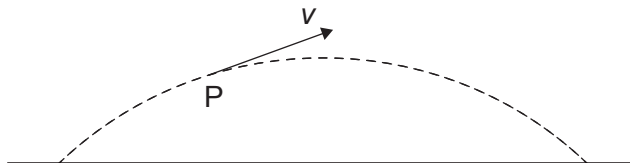


## PAPER 1 / QUESTION 48

8822-6501 / original question 5

48. The diagram shows the trajectory of a projectile and the velocity  $v$  of the projectile at point P in its trajectory. P is located before the projectile reaches the peak altitude. Air resistance acts on the projectile. The acceleration of the projectile at P is  $a$ .



What are the magnitudes of the horizontal component and the vertical component of the acceleration of the projectile at P?

|    | Horizontal component of $a$ | Vertical component of $a$           |
|----|-----------------------------|-------------------------------------|
| A. | zero                        | greater than $9.8 \text{ m s}^{-2}$ |
| B. | non-zero                    | greater than $9.8 \text{ m s}^{-2}$ |
| C. | zero                        | $9.8 \text{ m s}^{-2}$              |
| D. | non-zero                    | $9.8 \text{ m s}^{-2}$              |